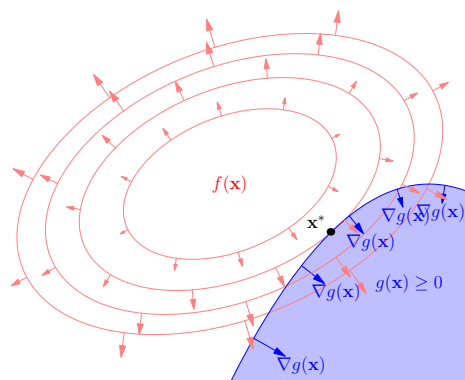
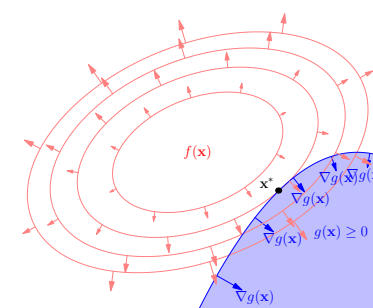




*Lagrangians, Inequalities, KKT, Linear Programming, Quadratic Programming, Duality*

1. **Constrained Optimisation**
2. Inequalities
3. Duality



## Optimisation with Constraints

- There are a number of important applications where we wish to minimise an objective function subject to inequality constraints
- A prominent example of this is support vector machines
- More generally there are a large number of kernel models that involve constraints
- However, constraints are ubiquitous in machine learning (e.g. in Wasserstein GANs)

## Solving Constrained Optimisation Problems

- Suppose we have a problem

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{subject to } g(\mathbf{x}) = 0$$

- A standard procedure is to define the Lagrangian

$$\mathcal{L}(\mathbf{x}, \alpha) = f(\mathbf{x}) - \alpha g(\mathbf{x})$$

where  $\alpha$  is known as a Lagrange multiplier

- In the extended space  $(\mathbf{x}, \alpha)$  we have to solve

$$\max_{\alpha} \min_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \alpha)$$

## Conditions on Optimum

- The optimisation problem is

$$\max_{\alpha} \min_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \alpha) \quad \text{where} \quad \mathcal{L}(\mathbf{x}, \alpha) = f(\mathbf{x}) - \alpha g(\mathbf{x})$$

- Assuming differentiability

$$\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \alpha) = \nabla_{\mathbf{x}} f(\mathbf{x}) - \alpha \nabla_{\mathbf{x}} g(\mathbf{x}) = 0$$

$$\frac{\partial \mathcal{L}}{\partial \alpha} = -g(\mathbf{x}) = 0$$

- The second condition is just the constraint
- But what about first condition:  $\nabla_{\mathbf{x}} f(\mathbf{x}) = \alpha \nabla_{\mathbf{x}} g(\mathbf{x})$ ?

## Note on Gradients

- Note that for any function  $f(\mathbf{x})$  we can Taylor expand around  $\mathbf{x}_0$

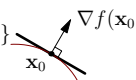
$$f(\mathbf{x}) = f(\mathbf{x}_0) + (\mathbf{x} - \mathbf{x}_0)^T \nabla_{\mathbf{x}} f(\mathbf{x}_0) + \frac{1}{2} (\mathbf{x} - \mathbf{x}_0)^T \mathbf{H} (\mathbf{x} - \mathbf{x}_0) + \dots$$

where  $\mathbf{H}$  is a matrix of second derivative known as the Hessian

- If we consider the set of points perpendicular to  $\nabla_{\mathbf{x}} f(\mathbf{x}_0)$  which go through  $\mathbf{x}_0$  (the tangent plane), these will have values

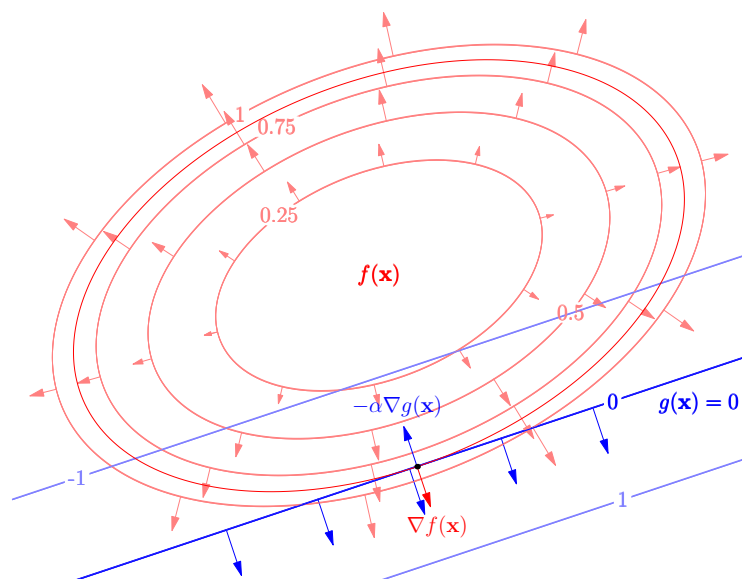
$$\{ \mathbf{x} | (\mathbf{x} - \mathbf{x}_0)^T \nabla f(\mathbf{x}_0) = 0 \}$$

$$f(\mathbf{x}) = f(\mathbf{x}_0) + O(\|\mathbf{x} - \mathbf{x}_0\|^2)$$



thus  $\nabla_{\mathbf{x}} f(\mathbf{x})$  is always orthogonal to the contour lines

## Constrained Optima



## Example

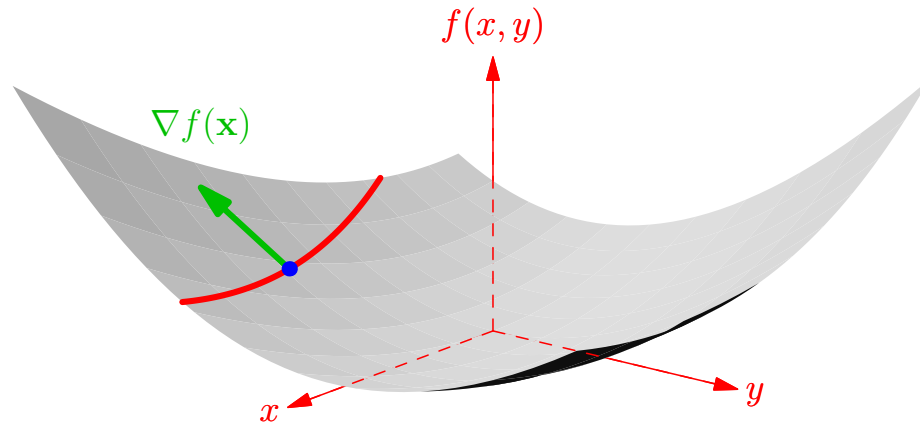
- Minimise  $f(\mathbf{x}) = x^2 + 2y^2 - xy$
- Subject to  $g(\mathbf{x}) = x - 2y - 3 = 0$
- Writing  $\mathcal{L} = f(\mathbf{x}) - \alpha g(\mathbf{x})$
- Condition for minima is  $\nabla_{\mathbf{x}} \mathcal{L} = 0$

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \begin{pmatrix} 2x - y \\ -x + 4y \end{pmatrix} = \alpha \nabla_{\mathbf{x}} g(\mathbf{x}) = \alpha \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

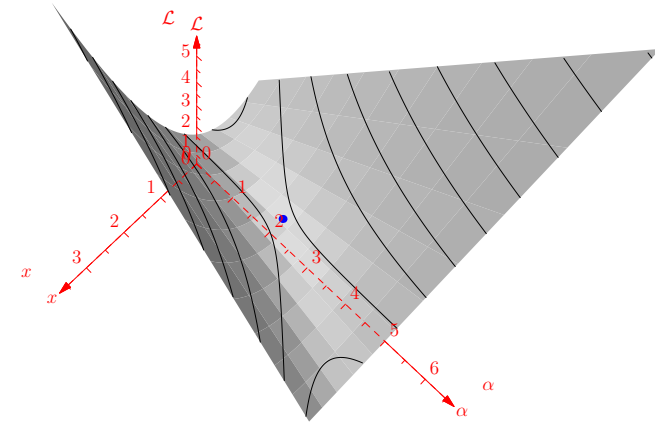
$$\text{and } \frac{\partial \mathcal{L}}{\partial \alpha} = -g(\mathbf{x}) = -x + 2y + 3 = 0$$

- Solving simultaneous equations gives minima at  $(x, y) = (\frac{3}{4}, -\frac{9}{8})$  with  $\alpha = \frac{21}{8}$

## Surface



## Saddle-Point $y = -9/8$



## Multiple Constraints

- Given an optimisation problem with multiple constraints

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{subject to } g_k(\mathbf{x}) = 0 \text{ for } k = 1, 2, \dots, m$$

- We introduce multiple Lagrange multipliers

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\alpha}) = f(\mathbf{x}) - \sum_{k=1}^m \alpha_k g_k(\mathbf{x})$$

- The condition for an optima is  $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\alpha}) = 0$  which implies

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \sum_{k=1}^m \alpha_k \nabla_{\mathbf{x}} g_k(\mathbf{x})$$

plus the original constraints  $\frac{\partial \mathcal{L}(\mathbf{x}, \boldsymbol{\alpha})}{\partial \alpha_k} = -g_k(\mathbf{x}) = 0$

## Example

- Minimise  $f(\mathbf{x}) = x^2 + 2y^2 + 5z^2 - xy - xz$  subject to  $g_1(\mathbf{x}) = x - 2y - z - 3 = 0$  and  $g_2(\mathbf{x}) = 2x + 3y + z - 2 = 0$

- Writing  $\mathcal{L}(\mathbf{x}, \boldsymbol{\alpha}) = f(\mathbf{x}) - \alpha_1 g_1(\mathbf{x}) - \alpha_2 g_2(\mathbf{x})$

- Condition for minima is  $\nabla_{\mathbf{x}} \mathcal{L} = 0$  or  $\nabla_{\mathbf{x}} f(\mathbf{x}) = \sum_{k=1}^2 \alpha_k \nabla_{\mathbf{x}} g_k(\mathbf{x})$

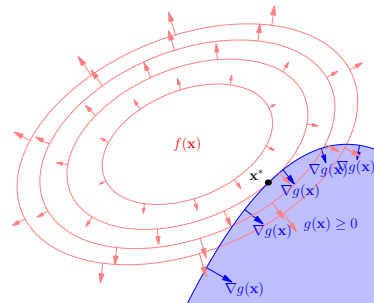
$$\begin{pmatrix} 2x - y - z \\ -x + 4y \\ 10z - x \end{pmatrix} = \alpha_1 \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

and  $\frac{\partial \mathcal{L}}{\partial \alpha_i} = -g_i(\mathbf{x}) = 0$

- Solving simultaneous equations gives minima at  $(\frac{37}{20}, -\frac{11}{20}, -\frac{1}{20})$  with  $\alpha_1 = 3$  and  $\alpha_2 = \frac{13}{20}$

## Outline

1. Constrained Optimisation
2. **Inequalities**
3. Duality



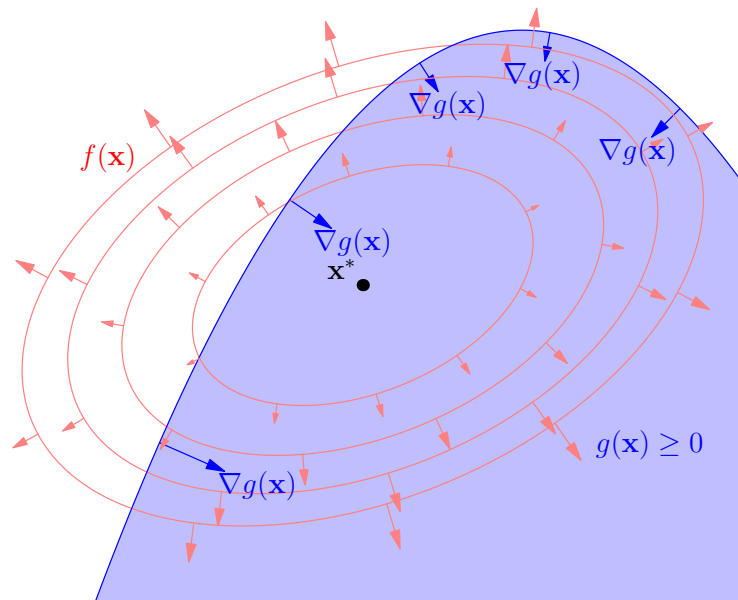
## Inequality Constraints

- Suppose we have the problem

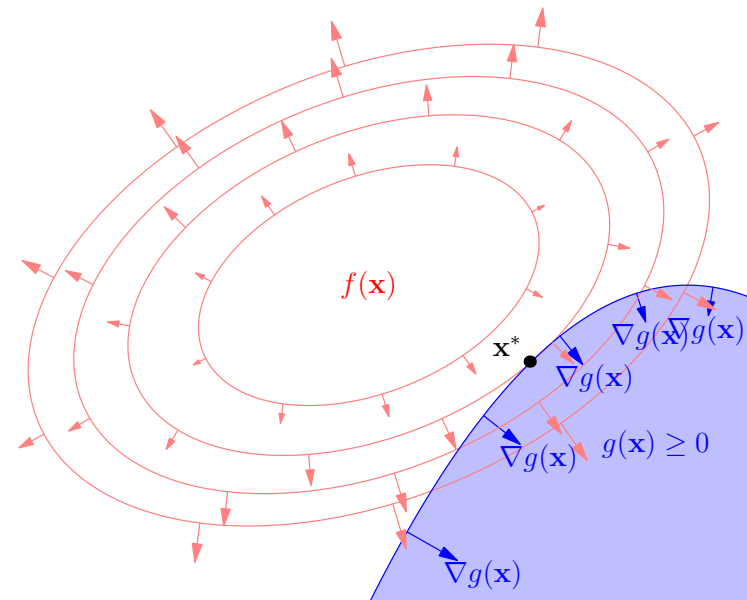
$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{subject to } g(\mathbf{x}) \geq 0$$

- Looks much more complicated, but
- Only two things can happen
  - ★ Either a minimum,  $\mathbf{x}^*$ , of  $f(\mathbf{x})$  satisfies  $g(\mathbf{x}^*) > 0$ 
    - \* We then have an unconstrained optimisation problem
  - ★ Otherwise, it satisfies  $g(\mathbf{x}^*) = 0$ 
    - \* We have a constrained optimisation problem

### Inside Region



### On the Boundary



## KKT Conditions

- To minimise  $f(x)$  subject to  $g(x) \geq 0$

$$\mathcal{L}(x, \alpha) = f(x) - \alpha g(x)$$

- Then  $\nabla_x \mathcal{L} = 0$  or

$$\nabla_x \mathcal{L} = \nabla_x f(x) - \alpha \nabla_x g(x) = 0$$

- where either
  - $\alpha = 0$  and the solutions in the interior
  - $\alpha > 0$  and  $g(x) = 0$ , i.e. the solution is on the boundary
- These conditions are known as the Karush-Kuhn-Tucker conditions

## Many Inequalities

- Given the problem

$$\min_x f(x) \quad \text{subject to } g_k(x) \geq 0 \text{ for } k = 1, 2, \dots, m$$

- We introduce multiple Lagrange multipliers

$$\mathcal{L}(x, \alpha) = f(x) - \sum_{k=1}^m \alpha_k g_k(x)$$

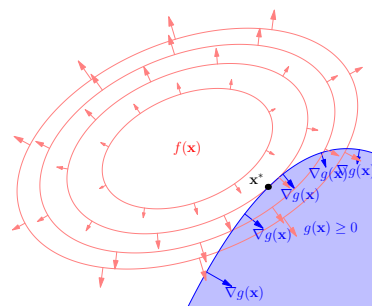
- The condition for an optima is

$$\nabla_x f(x) = \sum_{k=1}^m \alpha_k \nabla_x g_k(x)$$

- Plus the constraints that either  $\alpha_k = 0$  or  $\alpha_k > 0$  and  $g_k(x) = 0$

## Outline

- Constrained Optimisation
- Inequalities
- Duality**



## Solving the Lagrangian for $x$

- Consider minimising a function  $f(x)$  subject to a set of constraints  $g_i(x) = 0$  or  $g_i(x) \leq 0$
- We can consider this a double optimisation problem

$$\max_{\alpha} \min_x \mathcal{L}(x, \alpha) = \max_{\alpha} \min_x \left( f(x) + \sum_i \alpha_i g_i(x) \right)$$

where there would be constraints on  $\alpha_i$  if we had an inequality constraint

## Dual Problem

- If  $f(x)$  and  $g_i(x)$  are simple we can sometimes find a set of variables  $x^*(\alpha)$  that minimises the Lagrangian

$$\nabla_x \mathcal{L}(x^*(\alpha), \alpha) = 0$$

- This leaves us with the **dual problem**

$$\max_{\alpha} \mathcal{L}(x^*(\alpha), \alpha)$$

- If we had an inequality constraint  $g_i(x) \geq 0$  then we would have the additional constraint in the dual problem  $\alpha_i \geq 0$

## Linear Programming Example

- Suppose we eat potatoes and rice and we want to ensure that we get enough vitamin A and C

|           | Potatoes | Rice | Daily Requirement |
|-----------|----------|------|-------------------|
| Vitamin A | 3        | 5    | 20                |
| Vitamin C | 5        | 2    | 24                |
| Price     | 5        | 4    |                   |

- We want to buy  $P$  kg potatoes and  $R$  kg of rice as cheaply as possible subject to fulfilling our vitamin requirement

$$\min_{P,R} 5P + 4R$$

$$\text{subject to } P, R \geq 0, \quad 3P + 5R \geq 20 \quad \text{and} \quad 5P + 2R \geq 24$$

## Linear Programming

- In linear programming we minimise a linear objective function  $c^T x$  subject to linear constraints  $g(x) = Mx - b = 0$  (or  $g(x) \geq 0$ )
- The Lagrangian becomes

$$\mathcal{L}(x, \alpha) = c^T x - \alpha^T (Mx - b)$$

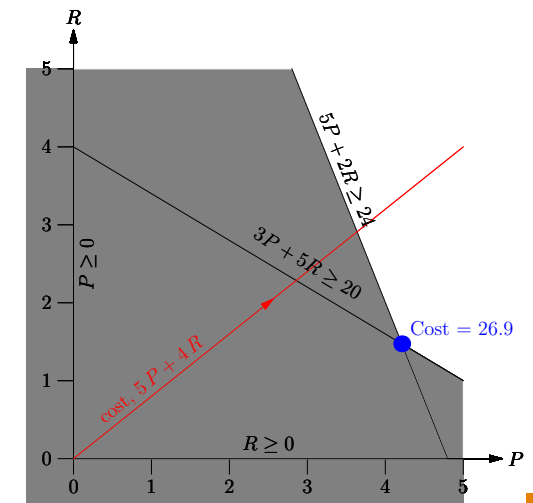
- An equivalent way of writing the Lagrangian is

$$\mathcal{L}(x, \alpha) = b^T \alpha - x^T (M^T \alpha - c)$$

- An entirely equivalent interpretation is that we maximise an objective function  $b^T \alpha$  subject to constraints  $M^T \alpha - c = 0$  (or  $M^T \alpha - c \leq 0$ )

## Linear Programming

- Minimise  $5P + 4R$
- Subject to
  - $3P + 5R \geq 20$
  - $5P + 2R \geq 24$
  - $P, R \geq 0$



## Lagrangian

- We can write the problem as a Lagrange problem

$$\min_{P,R} \max_{A,C} 5P + 4R - A(3P + 5R - 20) - C(5P + 2R - 24) \blacksquare$$

- subject to  $P, R, A, C \geq 0$   $\blacksquare$
- $A$  and  $C$  are Lagrange multipliers for vitamin A and C  $\blacksquare$
- We can rearrange the Lagrangian to obtain

$$\max_{A,C} \min_{P,R} 20A + 24C - P(3A + 5C - 5) - R(5A + 2C - 4) \blacksquare$$

## Dual Problem

- The Lagrangian

$$\max_{A,C} \min_{P,R} 20A + 24C - P(3A + 5C - 5) - R(5A + 2C - 4) \blacksquare$$

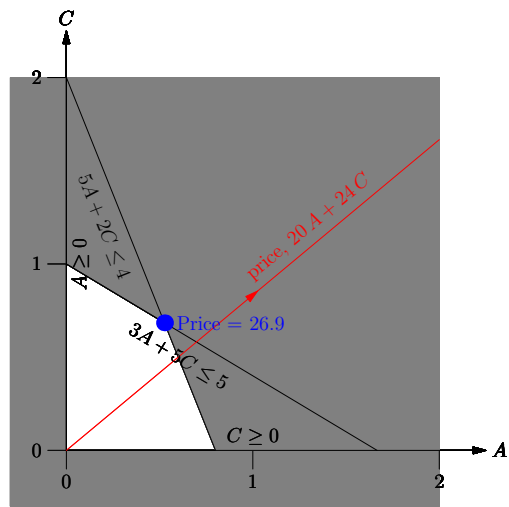
leads to the dual problem

$$\begin{aligned} & \max_{A,C} 20A + 24C \\ & \text{subject to } 3A + 5C \leq 5 \quad 5A + 2C \leq 4 \quad A, C \geq 0 \blacksquare \end{aligned}$$

- Consider someone selling vitamins A and C. They want to maximise the price of vitamins A and C, but their prices cannot exceed the price of the vitamins in potatoes or rice  $\blacksquare$

## Dual Linear Programme

- Maximise  $20A + 24C$
- Subject to
  - ★  $3A + 5C \leq 5$
  - ★  $5A + 2C \leq 4$
  - ★  $A, C \geq 0$   $\blacksquare$



## Why?

- Why are we bothered about translating one linear programme into another?  $\blacksquare$
- Sometime one form is massively easier to solve than the other  $\blacksquare$
- This is because the first linear programme depends on the dimensionality of  $x$  while the second linear programme depends on the number of constraints (or dimensionality of  $\alpha$ )  $\blacksquare$
- This is important, for example, in Wasserstein GANs  $\blacksquare$

## Quadratic Programming

- A quadratic programme involves minimising a quadratic function  $x^T Q x$  (with  $Q \succ 0$ ) subject to linear constraints  $Mx = b$  (or  $Mx \leq b$ )

- We can define the Lagrangian

$$\mathcal{L}(x, \alpha) = x^T Q x - \alpha^T (Mx - b)$$

- Where the solution is given by  $\max_{\alpha} \min_x \mathcal{L}(x, \alpha)$
- If the constraints are inequality constraints then  $\alpha_i \geq 0$

## Solution to Quadratic Programming Problem

- Using

$$\mathcal{L}(x, \alpha) = x^T Q x - \alpha^T (Mx - b)$$

- Then

$$\nabla_x \mathcal{L}(x, \alpha) = 2Qx - M^T \alpha$$

- So  $\nabla_x \mathcal{L}(x, \alpha) = 0$  implies

$$x^* = \frac{1}{2} Q^{-1} M^T \alpha$$

## Dual Quadratic Programming Problem

- Substituting  $x^* = \frac{1}{2} Q^{-1} M^T \alpha$  into

$$\mathcal{L}(x, \alpha) = x^T Q x - \alpha^T (Mx - b)$$

- We get the dual problem

$$\max_{\alpha} -\frac{1}{4} \alpha^T M Q^{-1} M^T \alpha + \alpha^T b$$

- If the constraints were inequality constraints then we have  $\alpha_i \geq 0$
- We have exchanged one quadratic programme for another, but sometimes that very useful (e.g. SVMs)

## Lessons

- A useful tool for performing constrained optimisation is the introduction of Lagrange multipliers
- This is particularly useful for problems with unique solutions (it will work when there are multiple solutions, but finding many saddle points is a pain)
- For inequality constraints we need to satisfy KKT conditions
- For simple situations (linear and quadratic programming) we can eliminate the original variables to obtain the dual problem